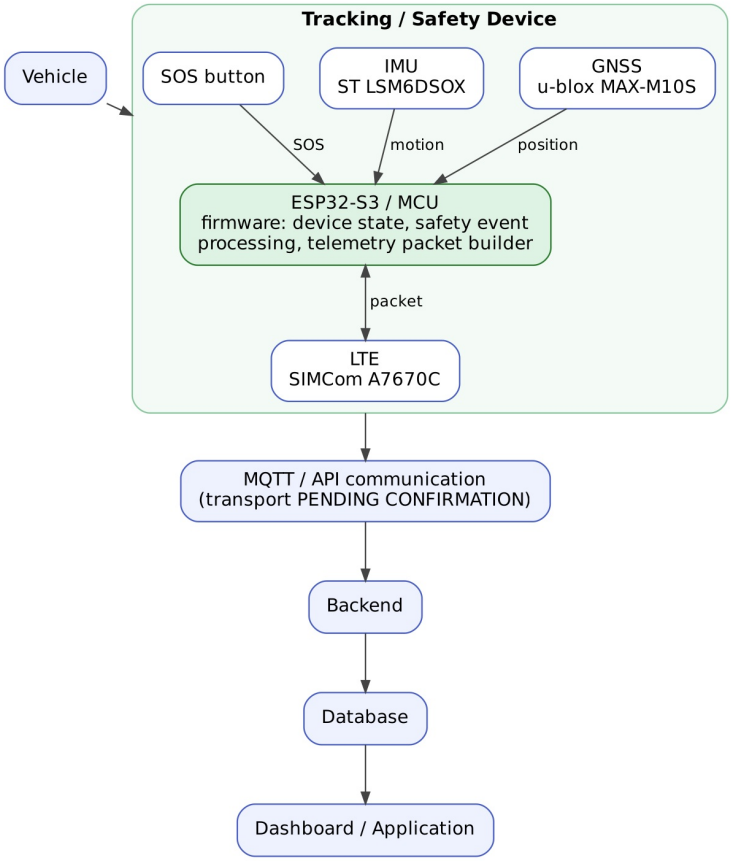
Week 2 - Day 1 (Monday): Firmware Architecture

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> Draft based on my current understanding. Anything I am not sure about is marked **Pending Confirmation**. Company-internal, please do not share outside the team.

1. Where the ESP32-S3 fits



The ESP32-S3 is the brain of the tracking/safety device. It sits between the sensors and the cloud:

1. It reads position data from the GNSS module and motion data from the IMU.
2. It reads the SOS button and runs the safety event processing.
3. It keeps track of the device state (starting, searching for GPS, connecting, operational and so on).
4. It collects everything into a telemetry packet and gives it to the LTE module, which sends it through MQTT or an API to the backend. The backend stores it in the database and the dashboard/application shows it.

Whether the data goes by MQTT or by an API is **Pending Confirmation**.

2. Approved components I studied

Component	Role in the device	What I found in the datasheet / product page	Pending Confirmation
ESP32-S3	Main MCU:	Dual-core MCU with Wi-Fi and	Pin

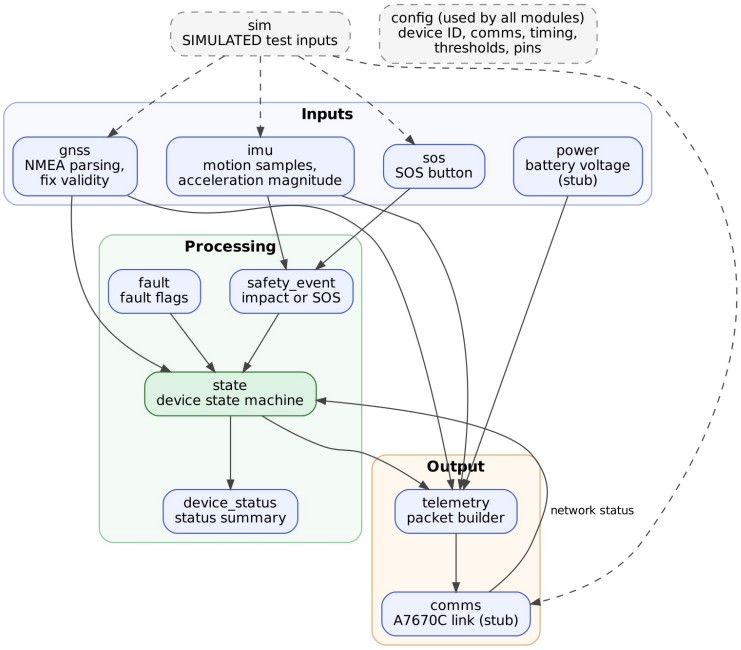
	runs the firmware, reads sensors, builds telemetry, talks to the modem	Bluetooth LE, 3.3 V logic (Week 1 study)	assignments
SIMCom A7670C	LTE Cat-1 cellular module that sends the data	Supply 3.4 to 4.2 V (3.8 V recommended). Peak current 2.8 A during GSM transmit bursts. UART default 115200 baud, **1.8 V logic**. PWRKEY low for at least 50 ms to power on, UART ready about 11 s later. GNSS is not listed as a feature in the hardware design document I read	Level shifting, power supply for the peak current, SIM/APN, MQTT or API
u-blox MAX-M10S-00B	GNSS (GPS) receiver	VCC and V_IO 2.7 to 3.6 V (3.0 V typical). UART default 9600 baud, 8N1. I2C (DDC) address 0x42, up to 400 kbit/s. Default NMEA 4.10 output: GGA, GLL, GSA, GSV, RMC, VTG, TXT. GPS, Galileo, GLONASS and BeiDou. Up to 10 Hz with several constellations. About 6 to 8 mA while tracking	UART or I2C, baud rate, update rate
ST LSM6DSOX	6-axis motion sensor (accelerometer + gyroscope)	I2C and SPI interfaces, supply 1.71 to 3.6 V	I2C or SPI, address, measurement ranges and data rate (to be read from the datasheet)
TI LMR33630-Q1	Automotive buck converter: steps the vehicle supply down	3.8 to 36 V input, 3 A output, AEC-Q100 grade 1 (-40 to +125 C ambient), versions for 400 kHz, 1.4 MHz or 2.1 MHz switching	Which version, output voltage setting
TI BQ24075	Single-cell Li-ion charger with power-path management	Operating input 4.35 to 6.4 V (absolute maximum 28 V). Charge current up to 1.5 A. 4.2 V charge voltage. Power-path (DPPM) powers the system and charges the battery at the same time. Charging/done and power-good status	System output voltage, charge current, whether the ESP32-S3 monitors the status pins
3400 mAh 18650 Li-ion	Backup power	Single cell, 4.2 V when full, about 3.6 to 3.7 V nominal	Protection circuit, required backup time

Notes: the MAX-M10S values come from an advance-information datasheet (revision R02), so they may change in a later revision. Sources are listed at the end of this document.

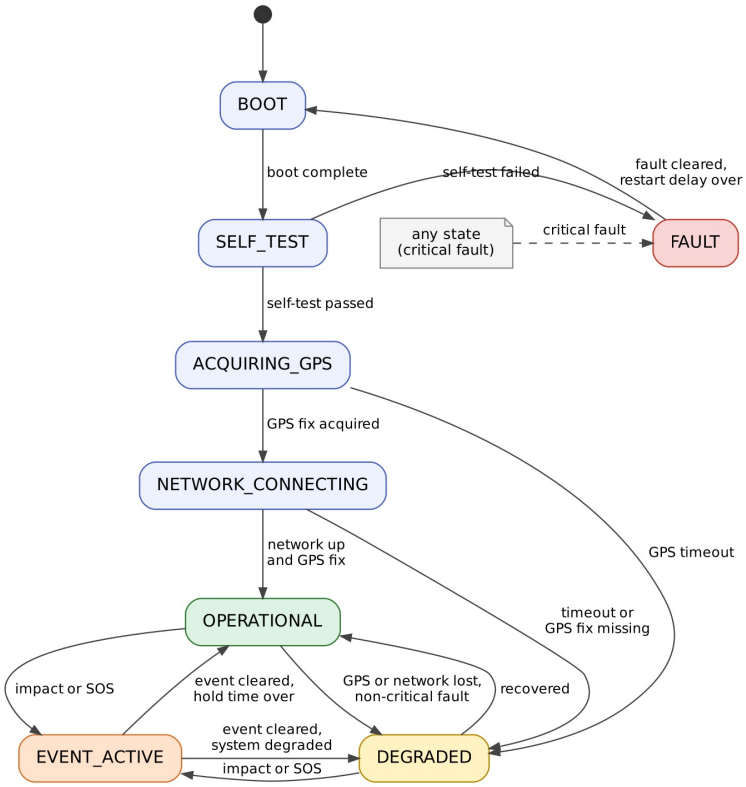
3. Firmware modules and what each one does

Module	One-line responsibility
System initialisation (main sketch)	Starts the modules in order, runs the main loop, calls the state machine and sends telemetry
config	Keeps all settings (device ID, intervals, timeouts, thresholds, pins, communication settings) in one place
gnss	Receives NMEA text from the u-blox MAX-M10S, checks the checksum, decodes it and reports a fix only when it is valid and recent
imu	Provides accelerometer and gyroscope samples and the acceleration magnitude; the LSM6DSOX hardware read is kept separate from the test-value source

comms	Manages the A7670C cellular link and sends the packets (stub for now)
safety_event	Decides whether the IMU data or the SOS button counts as an event (placeholder rule)
sos	Reads the SOS button
state	Decides which state the device is in and when it changes
device_status	Keeps one summary of the module health for logs and telemetry
fault	Keeps the fault flags and says whether a fault is critical
telemetry	Collects the readings into one packet and marks unavailable values as null
power	Reads the battery voltage (stub, value unavailable for now)
sim	Test only: feeds recorded NMEA data and test values so the firmware can be demonstrated without hardware



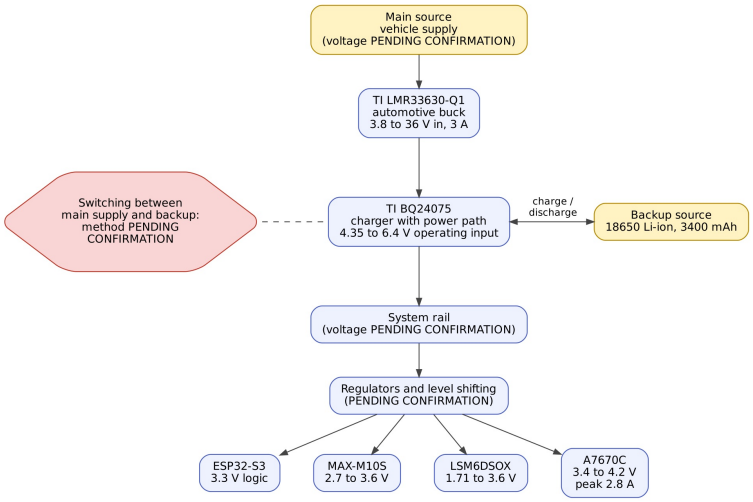
4. Device-state diagram



State	Meaning	Entered when
BOOT	Start-up, modules are initialised	Power on, or restart after a fault
SELF_TEST	Quick check of the configuration and module readiness	After BOOT
ACQUIRING_GPS	Waiting for the first valid GNSS fix	Self-test passed
NETWORK_CONNECTING	GPS fix is available, waiting for the cellular network	GPS fix acquired
OPERATIONAL	Normal operation: GPS fix and network up, regular telemetry	Network connected with a fix
EVENT_ACTIVE	An impact is suspected or SOS was pressed; telemetry is sent faster	Event detected
DEGRADED	Device still runs but something is missing (GPS, network or a non-critical fault)	Timeout, signal lost or non-critical fault
FAULT	Critical fault; telemetry stops and the device waits to restart	Critical fault from any state

The exact timeouts and the restart behaviour after FAULT are **Pending Confirmation**.

5. Power overview



- **LMR33630-Q1** steps the vehicle supply down to a lower rail and is automotive grade.
- **BQ24075** charges the 18650 cell and manages the power path, so the system can run from the input or from the battery.
- **18650 battery** keeps the device running when the vehicle supply is lost.

As advised, exact voltage values, charging current, the final power connections and the **method of switching between the main supply and the backup battery are Pending Confirmation**. I am not designing the final circuit.

6. Observations to raise (from the datasheets)

1. The A7670C UART uses 1.8 V logic, while ESP32-S3 pins are 3.3 V. Level shifting will be needed (PCB team).
2. The A7670C can draw 2.8 A peaks, while the BQ24075 limit is 1.5 A. How the modem is powered needs confirmation (PCB team).
3. The BQ24075 operating input is 4.35 to 6.4 V, so a buck stage must come before it. The intermediate voltage is Pending Confirmation.
4. The MAX-M10S supply and I/O are 2.7 to 3.6 V, so it can share the 3.3 V rail with the ESP32-S3 (to be confirmed by the PCB team).
5. The LMR33630-Q1 product page shows two different maximum output voltages (1 to 24 V in the features and 34 V in the table). I will check the datasheet table.

All open questions are collected in `Pending_Confirmation.pdf`.

Sources

- u-blox MAX-M10S datasheet (UBX-20035208): https://cdn.sparkfun.com/assets/7/5/9/a/a/MAX-M10S_DataSheet_UBX-20035208.pdf
- SIMCom A7670 Series Hardware Design V1.00: https://www.elecrow.com/download/product/CRT01260S/A7670_Series_Hardware_Design_V1.00.pdf
- TI BQ24075T product page: https://www.ti.com/product/BQ24075T
- TI LMR33630-Q1 product page: https://www.ti.com/product/LMR33630-Q1
- LSM6DSOX supply range and interfaces: retailer listing (RS Components), to be confirmed in the ST datasheet